The Settled-Gate Spacecraft Fault-Tolerant-Control Benchmark: An Honest Protocol for Comparable Learned and Classical FTC

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Abstract

Recent learned fault-tolerant control (FTC) of spacecraft attitude is reported almost entirely in simulation, on author-chosen fault sets, and against a *transient-minimum* success criterion that a trajectory need only touch once. On the same controllers, that criterion reads ∼94% where an honest steady-state gate reads ∼60%, and the gap is limit-cycling that a touch-once metric hides. We release a reproducible benchmark that closes this gap and gives the subfield a single shared standard of evidence. A controller *settles* a fault when its pointing error is held within 0.2° over a final-dwell window, scored on the *true* simulator state and never on a corrupted measurement. Faults are drawn from a *structurally held-out* taxonomy over inertia, gain magnitude, sign pattern, and additive bias, so a high score is generalization rather than an i.i.d. resample of training. Each cell is 500 episodes with Wilson 95% confidence intervals, and the environment is a pinned, pip-installable 6-DOF Basilisk simulator. We supply an audited evaluation harness so that any policy is a drop-in, together with a reference leaderboard spanning fault-unaware, classical-adaptive, two literature adaptive laws, end-to-end RL, and a structured estimate-then-control design. The gate exposes one finding starkly. Constant additive actuator bias (GAIN+BIAS) is recovered by no classical, learned, or RL controller, and not by a privileged gain oracle either, because an integral-free law cannot null a constant bias; only a disturbance-observer-augmented design recovers it (59.4%). We release the benchmark so that future FTC results are comparable and the gate is shared.

1 Why the subfield needs a shared honest gate

A spacecraft that loses, reverses, or degrades an actuator must recover its own pointing between sparse ground contacts, often on the order of one short pass per orbit and sometimes far less when a deep-space asset competes for a saturated Deep Space Network [16]. The recovered state that matters operationally is not a fleeting dip toward the target during a transient; it is a *held* attitude that science instruments, communication antennas, and momentum management can all rely on for the remainder of the contact gap. Fault-tolerant attitude control is therefore a crowded and well-motivated subfield, and learned methods have entered it in force. Yet the field's empirical claims are difficult to compare against one another, and difficult to trust as predictors of held-state recovery, because the community has not converged on a common standard of evidence.

A reading of the 2024–26 learned-FTC literature surfaces three recurring evaluation habits that, in combination, inflate reported success and frustrate comparison. First, results are *sim-only*, which is appropriate for early-stage research but is rarely paired with a clearly pinned and reinstallable environment, so a number reported in one paper cannot be regenerated under another. Second, the fault set is *author-chosen* and frequently overlaps the training distribution, so a high score can be an in-distribution resample rather than evidence of generalization to the off-nominal regimes a real failure produces. Transformer- and distillation-based FTC for fixed-wing and rotary platforms, for instance, is typically demonstrated on fault magnitudes inside the adaptation envelope it was trained over [12, 11]. Third, and most consequentially, "recovery" is scored as a *transient minimum*, the pointing error dipping below a threshold at some instant rather than holding a steady state. A controller can satisfy a transient-minimum gate while limit-cycling indefinitely and never actually pointing, which is precisely the failure an operator cannot tolerate.

These habits are not malpractice; they are the natural local optima of a subfield without a shared gate, where each paper picks the metric, fault set, and testbed that best exhibit its own method. The cost is collective. Numbers are not comparable across papers, headline success rates systematically overstate held-state recovery, and the genuinely hard fault classes, the ones that separate a controller that recovers from one that merely grazes, are obscured rather than surfaced. The remedy is not a better single method but a better shared standard, a gate the whole subfield can adopt, on which an honest number is lower but means what it says.

1.1 Empirical case for the gate

The choice of gate materially changes the standings, and its effect is large and measurable. On the identical controllers and faults of Table 2, a transient-minimum gate reports $\sim$94% success while the settled gate of this benchmark reports $\sim$60%. The $\sim$34-point gap is not noise and not a tuning artifact; it is steady-state limit-cycling, in which the controller touches the target once and then oscillates, behavior the touch-once metric scores as a clean success and the settled gate, correctly, does not. A gap of this size on unchanged controllers is the quantitative case for the standard. A benchmark that does not hold the controller to a dwell cannot distinguish recovery from a graze, and any leaderboard built on such a metric ranks controllers by a quantity an operator does not care about. The settled gate is the rigorous default the subfield needs, and the size of the gap is the measure of how much an easier gate has been hiding.

2 The protocol

The benchmark fixes four design decisions, each chosen to make a reported number both honest and comparable. We state each decision with its rationale, because a standard is only as useful as the reasons others have to adopt it.

2.1 The settled, steady-state gate

A controller *settles* a fault when the pointing error stays within $\mathbf{0.2}^\circ$ (the science-pointing requirement) over the **final dwell window** (20 steps = 10 s), evaluated on the **true** simulator state rather than at a transient minimum the trajectory touches once, and never on a corrupted measurement. The dwell is what separates a held attitude from a graze, and it is the property an operator actually requires, since an instrument integrates over the window rather than at the single best instant. Scoring on the true state, not the measurement, is what keeps a sensor fault from being trivially "passed" by a controller that has merely been fooled into reporting success. An *operational* gate ($\leq 5^\circ$ held) is reported alongside, so that a controller which stabilizes coarsely but misses the science threshold is visibly distinguished from one that diverges. Each cell is n=50 held-out faults $\times$ 10 seeds = **500** episodes.

2.2 Wilson confidence intervals

A success rate over 500 binary episodes is a binomial proportion, and at the extremes that dominate an FTC leaderboard, near 0% and near 100%, the familiar normal-approximation (Wald) interval is badly miscalibrated and can even extend past $[0, 1]$. The Wilson score interval [2] is the appropriate small-and-skewed-proportion estimator, remains inside $[0, 1]$, and does not collapse to zero width at a perfect cell. Reporting the Wilson 95% interval on every cell makes the leaderboard's uncertainty explicit and prevents a one-episode difference between two near-perfect controllers from being read as a real separation. Calibrated interval reporting of this kind is now standard practice in adjacent machine-learning-with-guarantees work [14], and an FTC benchmark should be held to the same bar.

2.3 Structurally held-out faults

The fault model is $\tau_{\text{applied}} = g \odot \tau_{\text{cmd}} + b$ with body inertia scaled by f, applied to the command inside the loop. Faults are split so the TEST set lives in regions of $(f, |g|, \text{sign}, b)$ the controller never trains on (Table 1). This is a deliberately stronger condition than the usual i.i.d. train/test resample. The test faults are not merely unseen samples from the training distribution, they lie *outside its support* on every axis, so a high TEST score is genuine extrapolative generalization rather than memorization or interpolation. A fault is, by its nature, an off-nominal event, so a benchmark that only ever tests inside the nominal envelope measures the wrong thing. The actuator classes are SIGN ($g \in \{\pm1\}$), GAIN (continuous g), GAIN+BIAS (g plus a constant bias b), and TOTAL_LOSS (one axis g=0, uncontrollable and so shield-only). The sensor classes, scored on the true state through an observation corruptor, are a constant attitude bias, star-tracker dropout, and a stuck measurement.

Table 1: Structurally held-out split. TEST faults lie outside the training support on every axis, so a high TEST score cannot be an i.i.d. resample of training.

Axis	TRAIN	TEST (held out)
Inertia scale f	$[0.8, 2.0]$	$[0.7, 0.8) \cup (2.0, 2.3]$
Gain magnitude $\lvert g\rvert$	$[0.5, 1.5]$	$[0.3, 0.5) \cup (1.5, 2.0]$
Sign pattern	≤ 1 reversed axis	≥ 2 reversed axes
Additive bias b	$[-0.2, 0.2]$	$[-0.3, -0.2) \cup (0.2, 0.3]$

2.4 A pinned, reinstallable environment

A 6-DOF Basilisk attitude simulator [1] is installed as a prebuilt wheel (`pip install bsk`; `import Basilisk`, v2.10.2), so that the testbed any submitter runs is byte-for-byte the testbed the reference numbers were produced on. Attitude is in modified-Rodrigues-parameter (MRP) coordinates [3], the observation is $[\sigma(3), \omega(3)] \in \mathbb{R}^6$, and the action is three normalized torque commands $\in [-1, 1]^3$. The timestep is 0.5 s, an episode is 400 steps (200 s), the inertia is diag$(10, 8, 6)$ kg·m^2, the max torque is 0.2 N·m, the nominal PD gains are (k_p, k_d)=(0.2, 1.5), and the settled gate is 0.2° over the final 20 steps. Pinning the version and shipping the wheel is what turns a reported number from a claim into a fact a reader can independently reproduce; a benchmark whose environment drifts between installs cannot anchor a leaderboard.

2.5 The evaluation harness

A controller is a fresh-per-episode policy $\pi(\text{obs}) \to a \in [-1, 1]^3$. The harness owns the parts of the experiment a submitter must not be free to vary, namely it applies the fault to the command inside the loop, advances the pinned simulator, scores the settled gate on the true state, and computes the Wilson interval over the canonical 500-episode battery. The submitter supplies only the policy. This separation is the point of the harness design, since the most common way a benchmark number becomes incomparable is that two authors silently differ on the gate, the fault battery, or the scoring side, and the harness removes exactly those degrees of freedom.

```
from program import rollout, eval_harness
from program.fault_taxonomy import FaultClass, heldout_test_faults

def my_maker(fault): # fresh policy per (fault, seed), state resets
    return lambda obs: ... # your controller
faults = heldout_test_faults(FaultClass.GAIN, n=50, seed=7000) # TEST battery
res = eval_harness.evaluate(
    rollout.make_rollout_fn(my_maker), faults, list(range(10)))
print(res["settled_science"]["rate"],
      res["settled_science"]["wilson95"])
```

Sensor-fault evaluation passes a `make_obs_corruptor`. Tune only on TRAIN, and report the settled gate and Wilson CIs on the canonical TEST battery. Submit an entry as a pull request adding a producer plus its evidence JSON, with every leaderboard number re-derivable by re-running and comparing to the committed value.

3 Related work

3.1 Fault-tolerant attitude control

The classical foundations of spacecraft attitude dynamics and control are well established [3, 4], and fault-tolerant variants have a long lineage in adaptive and robust control [5]. The discrete control-direction problem, in which an actuator's sign is unknown or reversed, is classically addressed by Nussbaum-gain methods [6], with attitude-specific treatments handling actuator saturation [7]. These methods are theoretically grounded and we include faithful implementations as baselines, but they are typically validated by Lyapunov argument and a small set of illustrative simulations rather than by a held-out statistical battery, so their behavior across a broad off-nominal fault distribution and a steady-state gate has not been systematically reported.

3.2 Learned FTC and its evaluation habits

A recent and fast-growing line applies learned adaptation to FTC. Estimate-then-control designs infer a latent fault parameter online and feed an analytic law, in the lineage of Neural-Fly [10] and Rapid Motor Adaptation [9], and transformer- or distillation-based controllers for rotary- and fixed-wing platforms [11, 12]. This work is genuinely capable, but it inherits the three evaluation habits described in Section 1, since it is reported in simulation, on fault sets that overlap the adaptation envelope, and against transient or episode-averaged criteria rather than a held steady state. To our knowledge, RMA-style FTC has not been evaluated on spacecraft attitude across a structurally held-out fault taxonomy with a settled gate, which is the gap this benchmark is built to close. End-to-end reinforcement learning [13] is a natural alternative hypothesis, and we include a from-scratch recurrent policy precisely so the benchmark can test whether learning *capacity*, absent estimate-then-control structure, suffices.

3.3 Benchmarking and reproducibility as a methodology

The broader lesson that a subfield's progress is gated by the quality of its shared evaluation, and that a rigorous, regenerable benchmark with a pinned environment and a precise scoring rule is itself a contribution, is by now standard in machine learning. Rigorous benchmarks that fix the task, the data, and the metric so that reported numbers are comparable and reproducible have repeatedly reset expectations in their fields [15], and calibrated-uncertainty reporting has become an expected companion [14]. The contribution here is to bring that same discipline, a pinned testbed, a precise and operationally meaningful gate, structurally held-out evaluation, interval reporting, and one-command reproduction, to learned spacecraft FTC, where it has been largely absent.

4 Reference leaderboard

Table 2 (actuators) and Table 3 (sensors) are the reference results for this project, generated directly from the committed evidence. They are the identical tables used by the companion controller study, `\input` here verbatim so the benchmark note and the study can never disagree. The leaderboard's purpose is not to crown a winner but to demonstrate that the gate is discriminating. On an easy metric the rows would compress toward the top and the table would say little, whereas on the settled gate the rows spread out and the genuinely hard classes declare themselves (Fig. 1).

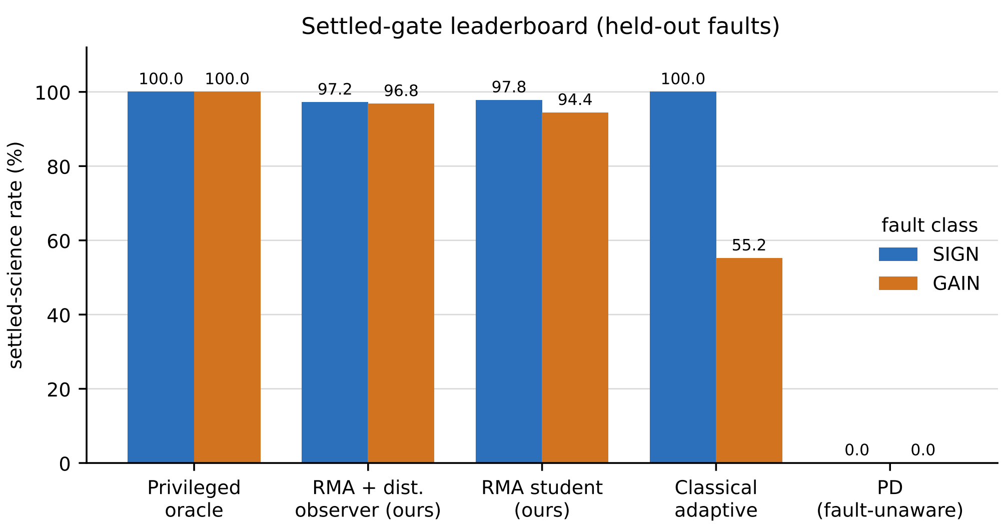

Figure 1: Headline standings on the two controllable continuous fault classes, ordered best to worst by combined settled-science rate. The settled gate spreads the controllers across the full range. The fault-unaware PD settles neither class, the classical adaptive law recovers SIGN but degrades on GAIN, and the RMA student approaches the privileged oracle on both. Every value is read from the same committed evidence as Table 2.

The leaderboard makes one finding unmissable that an easier gate would bury. **Constant additive actuator bias is 0% for every classical, learned, and end-to-end controller,**

Table 2: Definitive controller battery, reporting settled-science rate ($\leq 0.2°$ held over the final dwell) on the four structurally held-out fault classes (n=500/cell, same faults and gate across rows). **GAIN+BIAS is 0% for every classical, learned, and RMA controller, and for the privileged gain oracle, which is the integral-free architectural limit; the RMA student composed with the disturbance observer is the only deployable controller to recover it**, with no SIGN/GAIN regression. A dead axis (LOSS) is uncontrollable (shield-only) for all.

Controller	SIGN	GAIN	GAIN+BIAS	LOSS
PD (fault-unaware)	0.0%	0.0%	0.0%	0.0%
Classical adaptive (sign-ID)	100.0%	55.2%	0.0%	0.0%
ICL-adaptive [lit.]	82.2%	38.2%	0.0%	0.0%
Nussbaum-gain [lit.]	45.2%	3.2%	0.0%	0.0%
End-to-end GRU+PPO	0.0%	0.0%	0.0%	0.0%
RMA student, latched [WS1]	97.8%	94.4%	0.0%	0.0%
RMA + disturbance obs. [ours]	**97.2%**	**96.8%**	**59.4%**	0.0%
Privileged teacher (gain oracle)	100.0%	100.0%	0.0%	0.0%
Bias-feedforward oracle (g+b)	100.0%	100.0%	95.8%	0.0%

and for the privileged gain oracle. This is not a tuning failure but a structural one, and the benchmark is what makes it visible. An estimate-then-control law that infers the gain but applies an integral-free control cannot null a constant torque bias, since dividing the bias term by the gain estimate leaves a residual that no amount of gain accuracy removes; the privileged gain oracle, which knows the true gain exactly, therefore also sits at 0%. That the oracle fails alongside every learned and classical method localizes the difficulty to the control architecture rather than to estimation quality. Only the structured design augmented with a *disturbance observer*, which recovers the constant bias from the rigid-body dynamics and is self-correcting for gain-estimate error [8], recovers the class (59.4%, against a bias-feedforward upper bound of 95.8%), with no regression on SIGN or GAIN. A dead axis (LOSS) is uncontrollable for all controllers and is the runtime shield's responsibility, not the controller's. A leaderboard scored on a transient minimum would have reported this entire column near the top and hidden the one architectural fact most worth knowing.

Table 3: Sensor-fault regimes (deployed RMA, settled gate, scored on the TRUE state). A constant attitude bias is controllable within the 0.2° gate and offsets to exactly −bias beyond it (the sensor-side analog of actuator GAIN+BIAS); a lost or stuck attitude sensor removes position feedback (shield-only, the analog of TOTAL_LOSS). The privileged teacher matches the controller on every row, so these are fundamental sensor-regime limits, and a sensor bias is unobservable from the corrupted measurement alone (unlike the actuator bias).

Sensor fault	settled-science	final pointing	regime
Sensor bias 0.1°	96.4%	0.10°	controllable
Sensor bias 0.2°	45.2%	0.20°	controllable
Sensor bias 0.3°	0.0%	0.30°	shield-only
Sensor bias 0.5°	0.0%	0.50°	shield-only
Star-tracker dropout	0.0%	97.19°	shield-only
Stuck measurement	0.0%	85.56°	shield-only

The sensor side is classified the same way and exhibits the same controllable-versus-shield-only boundary. A constant attitude-measurement bias is controllable within the gate and offsets to exactly −bias beyond it, the sensor-side analog of the actuator GAIN+BIAS wall, and unlike the actuator bias it is unobservable from the corrupted measurement alone, so it cannot be removed by a disturbance observer. A lost or stuck attitude sensor removes position feedback entirely and is a shield-only regime, the analog of TOTAL_LOSS. That the privileged teacher matches the controller on every sensor row indicates these are fundamental regime limits rather than controller deficiencies, which is exactly the kind of distinction a steady-state gate scored on the true state is able to draw.

5 Discussion

A shared honest gate changes what the subfield can say. Once recovery means a held steady state on faults outside the training support, scored on the true state with reported intervals, three things follow that an easier metric prevents. First, numbers become comparable, and a 59.4% on this gate means the same thing in any paper that adopts it, so results accumulate into a leaderboard rather than a collection of incomparable claims. Second, the hard classes surface instead of hiding. The GAIN+BIAS wall is invisible to a transient-minimum metric and obvious here, and surfacing it is what turned it from an unnoticed 0% into a posed problem with a disturbance-observer answer; a standard that hides its hardest cases cannot drive progress on them. Third, the burden of proof shifts in the right direction. Under a shared gate a high number is a strong claim a reader can independently reproduce from a clean checkout, rather than a figure contingent on an author's private metric and testbed.

We expect the immediate effect of adopting the gate to be that some published successes read lower, which is the intended correction rather than a defect. A 94% that was really a 60% was never a property of the controller; it was a property of the metric, and replacing it with a number that means held-state recovery is what lets the subfield trust its own leaderboard. The gate is deliberately strict so that the easy way to a high score, optimizing the metric rather than the behavior, is closed off. The contribution we are most committed to is therefore the gate and the harness, not any particular leaderboard position; the position is evidence that the gate discriminates, and the gate is the thing we are asking the subfield to share.

5.1 Limitations and scope

The benchmark is single-environment and seed-controlled, which is what makes it exactly reproducible but also means it is a protocol for comparable *simulation* evidence and a TRL-2-to-3 instrument, not a flight qualification. The actuator model is external-torque injection rather than a full reaction-wheel-momentum model, a fidelity upgrade we regard as the natural next version of the testbed rather than a change to the protocol. The 0.2° science gate, the 200 s horizon, and the fault-magnitude ranges are concrete and defensible choices for the rigid-body regime modeled here, and a different mission class would re-instantiate the same protocol with its own such constants; the protocol, not the particular numbers, is the portable contribution. Finally, the leaderboard reported here is one project's reference set, and the submission rule exists precisely so that the gate, once shared, is populated by the community rather than by us.

6 Release and reproduction

The benchmark package (`benchmark/BENCHMARK.md` and the auto-generated leaderboard) pins the gate, taxonomy, environment, harness, and submission rule. The leaderboard producers each write verifiable evidence JSON, headline numbers are re-checked from a clean checkout by a single reproduce script, and any single number is re-derivable with a verify-number utility that re-runs it and compares to the committed value. We release the package so that every learned-FTC paper that adopts the settled gate is directly comparable to every other. The gate, not the leaderboard position, is the contribution.

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